

Unit-6 : Ethics & Responsible AI

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Why Ethics in AI Matters

- AI systems affect billions of people daily
- Decisions in healthcare, finance, justice, hiring
- Potential for both tremendous benefit and harm

Key Challenges

- Algorithmic bias and discrimination
- Privacy violations and data misuse
- Lack of transparency and accountability

Understanding Bias in AI

What is AI Bias?

Systematic and unfair discrimination by AI systems against certain individuals or groups, leading to prejudiced outcomes.

➤ **Types of Bias:**

- **Historical Bias:** Training data reflects past discriminatory practices
- **Representation Bias:** Certain groups underrepresented in data
- **Measurement Bias:** Incorrect feature selection or labeling
- **Aggregation Bias:** One-size-fits-all model for diverse populations



Real-World Examples of AI Bias

Example 1: COMPAS Recidivism Algorithm

Used in US courts to predict reoffending risk. ProPublica found it was twice as likely to incorrectly flag Black defendants as high-risk compared to white defendants.

Example 2: Amazon Hiring Algorithm

Amazon's AI recruiting tool showed bias against women because it was trained on resumes submitted over 10 years (predominantly from men). It downranked resumes containing the word "women's" (e.g., "women's chess club").

Example 3: Facial Recognition Systems

MIT and Stanford study found major facial recognition systems had error rates up to 34% higher for darker-skinned women compared to lighter-skinned men, leading to wrongful arrests.

Fairness in AI Models

Fairness Definitions

Individual Fairness

Similar individuals should receive similar predictions

Group Fairness

Protected groups should have equal outcomes

Demographic Parity

Positive outcome rates equal across groups

Equalized Odds

Equal true/false positive rates across groups

Key Insight: Different fairness definitions can conflict with each other. Organizations must choose which definition best fits their ethical framework and use case.

Strategies to Mitigate Bias

Pre-Processing

- Diverse and representative datasets
- Data augmentation for underrepresented groups
- Careful feature selection

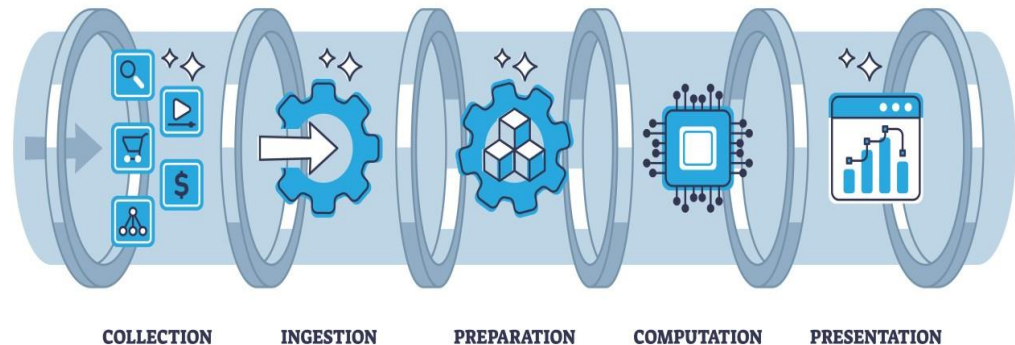
In-Processing

- Fairness constraints during training
- Adversarial debiasing techniques
- Regularization methods

Post-Processing

- Threshold adjustment by group
- Calibration techniques
- Reject option classification

DATA PIPELINE



Privacy in AI Systems

Privacy Concerns

- **Data Collection:** Massive amounts of personal data collected without informed consent
- **Re-identification:** Anonymized data can be re-identified using AI techniques
- **Inference Attacks:** Models can leak training data information
- **Surveillance:** AI enables mass surveillance and tracking

Example: Cambridge Analytica Scandal

Personal data of 87 million Facebook users was harvested without consent and used for political advertising. AI algorithms analyzed personality traits to target voters with personalized political messages.

Privacy-Preserving AI Techniques

Differential Privacy

Adds mathematical noise to data or queries to prevent identification of individuals while preserving statistical accuracy

Example: Apple uses differential privacy for keyboard predictions

Homomorphic Encryption

Performs computations on encrypted data without decrypting it

Example: Secure medical data analysis

Federated Learning

Trains models on decentralized data without transferring raw data to central servers

Example: Google's Gboard learns from user typing patterns locally

Secure Multi-Party Computation

Multiple parties jointly compute functions while keeping inputs private

Example: Financial institutions sharing fraud detection insights

Security Threats in AI

Adversarial Attacks

- **Evasion:** Slightly modified inputs fool models
- **Poisoning:** Malicious data injected during training
- **Model Extraction:** Stealing model parameters through queries

Example: Researchers added small stickers to stop signs that caused Tesla's autopilot to misclassify them as speed limit signs

Defense Strategies: Adversarial training, input validation, model ensemble methods, robust architecture design, continuous monitoring

Explainability & Transparency

The Black Box Problem

Complex AI models (especially deep neural networks) make decisions that humans cannot easily understand or explain.

Why It Matters

- Legal and regulatory compliance (GDPR "right to explanation")
- Building trust with users
- Debugging and improving models
- Ensuring accountability
- Detecting bias and errors

XAI Techniques

- **LIME:** Local interpretable model-agnostic explanations
- **SHAP:** Shapley additive explanations
- **Attention Maps:** Visualize what model focuses on
- **Feature Importance:** Rank influential features

Ethical Implications of AI Applications

Autonomous Weapons

AI-powered weapons that can select and engage targets without human intervention raise questions about accountability, proportionality, and the value of human life.

Social Media Algorithms

Recommendation systems can create echo chambers, spread misinformation, and negatively impact mental health, especially among youth.

Predictive Policing

Using AI to predict crime can perpetuate systemic biases, over-police minority communities, and create self-fulfilling prophecies.

Deepfakes & Synthetic Media

AI-generated fake videos and audio can be used for fraud, political manipulation, revenge porn, and destroying reputations.

Building Responsible AI: Best Practices

1. Ethics by Design

Integrate ethical considerations from the beginning of the development process

2. Diverse Teams

Include diverse perspectives in AI development and deployment

3. Impact Assessments

Conduct thorough assessments of potential harms before deployment

4. Continuous Monitoring

Regularly audit AI systems for bias, fairness, and performance

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